Promoting Student Learning in Support of a Farmer Research Network in Migori County

Report on a pilot study involving Kuria west primary school students

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Executive Summary

A pilot study was conducted in in Kurutiyange and Naora villages, Kuria West sub-county, Migori County, Kenya. The study's two objectives were (1) to explore whether primary school students can support their community's Farmer Research Network (FRN) by collecting higher quality data than their parents who are involved in practice-oriented agricultural research. The second was to pilot an example of practical, multi-faceted student learning modules that can help student develop the key skills and competencies for solving daily life challenges. The latter is the focus of Kenya's recently implemented CBC (Competency Based Curriculum). The pilot study was a collaborative effort supported by the Kenya Agroecology Hub for the agricultural aspects and by INNODEMS Kenya for the educational ones.

Our study involved 171 fifth-grade students and incorporated various stakeholders, including FRN farmer representatives, educational specialists, and local administrators and village chiefs. The study was divided into two phases: the first phase introduced students to basic research skills, and the second phase saw students conduct scientific experiments on the effectiveness of *Tephrosia vogelii* in controlling their household poultry's ectoparasites.

Through the implementation of the project over a 6 month period (2 school terms), students were exposed to valuable skills in agriculture, research, data collection, and interviewing. They learned about parasites on farm animals, as well how to formulate questionnaires, conduct interviews, and analyse data.

However, the pilot study's hypothesis that these particular primary school students could collect higher quality data than their parents was ultimately rejected. Several challenges, such as students being sent home for non-payment of school fees, poor handwriting, and the co-occurrence of various cultural events that kept students away from school and/or home, affected data quality. Additionally, we found inconsistencies in students' data, as well as false responses and misinterpretation of questions, which further compromised students' data quality. Still, despite the challenges experienced, we found evidence that the students gained much educational value from being part of the study. Hence, we recommend learning from the challenges encountered in the study to improve future involvement of students in FRN activities and school lessons that promote the acquisition of relevant life skills and competencies by Kenyan students.

Abbreviations

AE-HUB Kenya	Agroecology Hub Kenya
CBC	Competency-Based Curriculum
CBO	Community-Based Organization
FGM	Female Genital Mutilation
FRN	Farmers Research Network
IDEMS	Innovations in Development, Education and the Mathematical Sciences
INNODEMS	Innovations in Development, Education and the Mathematical Sciences Kenya
MEL	Monitoring, Evaluation and Learning
KCPE	Kenya Certificate of Primary Education

Introduction

Farmers' Research Network (FRN) is a research approach that puts farmers at the heart of agricultural research processes. The FRN approach involves networking groups of farmers from different geographic areas, who have a shared priority challenge, to test a shared set of agroecologically oriented treatments on their own farms. They compare these practices to their own current practice and evaluate whether any of the new practices are superior. The approach can rapidly reveal if technologies are effective vs ineffective. It also reveals if results are consistent across diverse environments or, if they are more context dependent, working in some environments and not others.

In Migori county, there is a vibrant farmer-led FRN where farmers have been engaged in testing the uses of a pesticidal plant, *Tephrosia vogelii*, to control locally important pests and diseases. The Rural Farmers Development Network (RFRN) is a registered community-based farmer organisation (CBO) made up of 200 farmers from six Migori county sub-counties. This RFRN is coordinated and implemented by leader farmers within the organisation, with technical support from the Kenya Agroecology Hub, based at Manor House Agricultural Centre. The organisation's leader farmers agree on a research topic, hash out and negotiate the details of the research design (with the research support team) and then train their fellow farmers how to conduct the experiment. They also do all the data collection.

In 2022 RFRN assessed Tephrosia's effectiveness at controlling external parasites in poultry (fleas, mites, ticks, and lice). During focus group discussions, after the research was concluded, members provided anecdotal evidence that *Tephrosia vogelii* worked. Moreover, the participating farmers were very excited by the results because they said that controlling the parasites had led to greater egg production and hatching in their flocks.

Over the past several years of experimentation various deficiencies in the data being recorded have been noted by the research support team. Data deficiencies have arisen as a result of several factors. Farmers are averse to experiencing any unnecessary loss so they quickly abandon any treatment that they observe to be ineffective Many farmers are too busy to consistently collect and record data. Other farmers do not see the value in data collection. All these scenarios result missing data. Still, poor data quality is not a problem for most RFRN farmers themselves since they rely on their own observations rather than written data to make their individual conclusions about the research outcomes, some of which have proven to be extremely useful to them. However, the data quality issues mean that farmers don't have the statistically validated evidence to convince the outside world about their findings. This report reviews a potential approach to improving FRN data quality through the involvement of primary school students from communities where there are active RFRN members.

A secondary aspect of the study was to test the value of including real life agricultural challenges and principles of agricultural experimentation in the school curriculum. Rural primary schools have struggled to implement the new competency-based curriculum (CBC), which was recently introduced. This curriculum encourages experiential learning (Kubai E., 2023). Experiential learning is at the heart of the Framework's Guiding Principle of Community Service Learning, where students work on real problems faced by their communities (KICD 2017, 19-20).

We saw that this project had the potential to encourage the learning that CBC is aiming for. The project developed lesson and learning activities cut across subjects, such as maths, English, science, agriculture, and Kiswahili. The use of diverse teaching approaches, including drawing, singing, and dramatization could also help improve student learning outcomes. Students could get hands-on learning experience in how to use the scientific method, collecting and understanding research data, and communicating their findings. This would help them develop four of the seven core competencies identified in the Curriculum Framework, notably communication and collaboration, critical thinking and problem-solving, citizenship, and learning to learn (KICD 2017). At the same time, the data collected by students could help to validate the FRN's findings for the outside world. Involvement of primary school students in FRN activities therefore seemed to have good potential to both improve FRN data outcomes and implement the experiential learning approach mandated by the CBC.

To test the potential of primary school students to contribute to quality data collection in FRNs, we undertook a 6-month pilot study in two primary schools in Migori County. The study sought to answer two questions: 1) Can primary school students collect high-quality data on the FRN research topics, and 2) Can introducing FRN activities in schools enhance student learning?

Methodology

The study was carried out in two primary schools in Kurutiyange and Naora villages, in Kuria west sub-county. These schools were chosen because there was a strong RFRN group in the villages, and members had enough supply of Tephrosia powder to be used by the students during the experiments. The team worked with primary school students as they transitioned from one academic year to the next. The first phase took place for 5th grade pupils during their last 3 months of classes and then continued for their first 3 months of 6th grade. These students were between 10 and 15 years old.

Several stakeholders were involved in project planning. The Kenya Agroecology Hub (AE Hub), hosted by Manor House Agricultural Centre, conceptualized the study, which then built up with the contributions of other actors. FRN farmer representatives provided important information and support for working within their communities. They connected the team to the local administrators, identified the specific schools to be involved, arranged accommodation, and reached out to parents and other FRN members within Kuria West Sub-county.

Educational specialists from two education-focused organizations, IDEMS (Innovations in Development, Education and the Mathematical Sciences) and INNODEMS (INNOvations in Development, Education and the Mathematical Sciences, the Kenyan affiliate of IDEMS) supported the educational aspects. The INNODEMS's Director visited the schools to help headteachers and teachers understand the relevance and methods to be used by the field team.

The others provided regular input on the development of teaching materials and learning approaches suitable for the students.

The study was carried out in two, three-month phases. Phase 1 supported students in developing basic research skills. In phase 2, students put the skills to practical use by participating in an experiment on external parasites in chicken.

Learning components

Four skills areas necessary for quality research were identified: 1) communication, 2) undertaking observations, 3) data recording, and 4) data analysis, visualisation, and interpretation. The teaching units focused on these skills. The first phase focused on skills 1-3, which form the basis for collecting good quality data. Students received training in general communication and observation skills through learn to play and play to learn activities and through more formal methods, including writing journals and conducting questionnaires and interviews.

In the second phase, students participated in the study on the effectiveness of *T. vogelii* in controlling poultry ectoparasites. This phase started with a questionnaire about rearing chickens at home that the students would complete with their parents. The questionnaire helped to determine students' homestead poultry flock sizes, and how much *T. vogelii* was to be distributed to the students. The experiment itself involved students observing their chickens parasite infestation level before and after *T. vogelii* treatment on a small sample of their poultry. The second phase also supported students in analysing and interpreting their data, the fourth skill set needed for good quality research.

Research tools and methods

(Appendix A)

During the project the team used several research tools to engage with students, their caregivers, and their teachers.

Baseline questionnaires

Students completed two baseline questionnaires to provide initial information about their household's chicken flock. Students completed an initial questionnaire themselves, then interviewed their caregivers to gather data on flock-rearing practices, chicken uses, and pests and diseases. The caregivers survey provided an opportunity for students to inform and engage their parents in the project. We hoped this would lead to students getting their parents' support for the project., especially for making observations of poultry behaviour and other flock dynamics while the students were at school.

Student-led flock experiments: questionnaires and holiday journaling

For the students' experiments on the effectiveness of *T. vogelii* in controlling poultry ectoparasites, students collected data through an intervention questionnaire. This form was completed over the six weeks of the student study and covered student observations of their flocks during the two weeks before application of *T. vogelii*, as well as their observations in the month following application. They recorded the type and number of parasites observed,

which part of the body was infested, and symptoms and behaviours exhibited by their chicken when sick. During the six weeks of the experiment, the students were asked to complement data collected in their intervention questionnaires by recording their observations of any abnormal chicken behaviours in a journal.

Home visits: semi-structured interviews

The team visited every students' home at the end of the first phase, and again at the end of the second phase. Caregivers were interviewed during these visits. The visits were also important for gauging if students were doing their homework and to gain insights about any difficulties they were having.

The semi-structured interviews also explored whether students received support from their caregivers, and what the caregivers' views were of *T. vogelii*. The second visit helped us to monitor the progress of the *T. vogelii* application experiment, assess the parents' role in making observations on the chickens, as well their opinion on the impact of the project on the student's learning.

Generally, home visits served to both strengthen parents' understanding and engagement in the research and to help us identify how we could improve the delivery of the project to increase its impact.

Teacher focus groups

Teachers were crucial in monitoring the quality of the study in terms of the accuracy and reliability of the data received, which related to student learning and the competency-based curriculum. Regular teacher meetings were therefore held in each school to get their feedback. Open discussions with the teachers helped us with monitoring the study's progress and impact and identification of potential improvements. We also closed the project with the schools and teachers involved by holding focus groups to reflect on the study and gather their end of project feedback.

Learning Materials and Approaches

Lesson plans were created in line with the CBC curriculum. (Appendix b- p37-61). There were four teaching units, namely communication skills, observation and recording, which had a total of 12 lesson activities. The lesson activities were initially designed in English, but Kiswahili was used during teaching and practical activities.

Outdoor activities used a play-to-learn approach, while hand lenses and foldscopes were used for hands-on activities. A Foldscope is a low-cost, portable microscope made from a single sheet of paper which offers higher magnification and imaging capabilities compared to hand lenses. With magnification up to 140x, LED illumination, and smartphone compatibility, the Foldscope is ideal for fieldwork, resource-limited environments, and education.

In one hands-on activity, the students were provided with 1 kg of *T. vogelii* powder and were shown how to prepare the extract for application. The students were also provided with tags to identify the chickens to be observed during the experiment.

Involving FRN members in the student learning process

During some lessons, two FRN members were actively involved. They assisted in instructing the students on how to prepare the *T. vogelii* extract, and demonstrated how and where to apply the *T. vogelii* treatment to the chickens. Additionally, during home visits, the FRN members acted as intermediaries and translators between the team and parents since some community members were not fluent in either Kiswahili or English.

Results and discussions

Phase 1 Gearing up for experimentation.

Resources developed

The team developed learning resources that were in line with the Competency Based Curriculum (CBC) (Appendix B, starting p37). The learning activities were compiled by integrating several related topics from different subjects within the CBC curriculum. For example, while animal rearing was a core activity drawn from Agriculture, skills such as counting, drawing graphs, presentation skills, and reading, were drawn from Mathematics, English, and Kiswahili, respectively. Every activity was designed to incorporate at least two skills from different subjects. Additionally, in attempts to mitigate the potential monotony and boredom associated with conventional teaching methodologies, the team adopted a "play to learn" approach. By integrating elements of play and interactive techniques into the instructional framework, we aimed to cultivate an engaging and enjoyable learning environment for the students involved.

The key objective behind the development of the lessons was to equip students with research-related skills. The lessons and related activities were broadly categorised into 4 units i.e.

1. Communication: This is aimed at developing the learners' communication skills through engagement in the charade and telephone games. Learners were meant to understand the importance of keenly observing behaviour, collecting information about the behaviour, and effectively communicating the behaviour.
2. Undertaking observations: This is aimed at developing the learners' observational skills through pictorial, video, and outdoor observations. The emphasis was on equipping students with knowledge and skills for making observations on behaviors and events in their natural settings.
3. Data recording: The is aimed at promoting the development of a wide range of written, oral, and visual skills involved in documenting and organizing information. Emphasis was on cultivating an understanding of the importance of recording observations, how to use the project's daily data tool, and improving the quality of recorded observations.
4. Data analysis, visualization and interpretation: This is aimed at building the student's ability to represent and understand real-world problems using charts, graphs, and tables. Learners were therefore to be taken through concepts of noticing patterns in the data and understanding emerging trends/patterns from the data.

The development of the learning resources was a continuous process that involved getting feedback from the teachers, researchers, education experts, and paying attention to what was working or not working during the facilitation of the learning sessions.

Students' reactions to the lessons

Students enjoyed the different approaches to learning that the team used for the lessons. These were very different than the regular learning model where students are passive recipients of their teacher's knowledge and where the emphasis is on "learning" by rote memorization and studying for standardized tests.

In this study, the learners had the opportunity to draw, sing, dramatize, and participate in small group discussions around the concepts being taught. The team also used videos and pictures shown using PowerPoint presentations to bolster students' understanding (Figure 1). These activities were not only fun but appeared to help students easily retain concepts related to chicken behaviour and chicken parasites.

Figure 1. Student at Naora Primary school leading a presentation on chicken behaviours while one of the study team members looks on.



Incorporation of foldscopes and hand lenses in their lesson activities aroused a curiosity and exploration mindset among the students. Students used hand lenses to observe various objects and insects (Figure 2). When combined with an activity involving them drawing their observations, it helped them to relate and understand more about animals and microorganisms. During the lessons, they were shown how to assemble and how to collect and prepare a sample that they used to make observations. These activities appeared to encourage an appreciation for the environment in the students.

Figure 2. Students from Kurutiyange (left) and Naora (right) primary schools during an outdoor lesson making observations on chicken using hand lenses





Student participation in the data collection activities

When we entered the classrooms at the beginning of the project, there were a total of 171 fifth graders on the rosters of the two schools. Fewer girls were present in both schools.

Student numbers varied a lot over the course of the 6 month study (Table 1). This was mainly because of absenteeism due to student being sent home when their school payments were in arrears.

Table 1: Fluctuation in student participation at different points in the study, based on the number of different data forms returned to the team.

	Student flock form	Caregiver form	Data form for Pre- *T. vogelii* application	Data form for post *T. vogelii* application
Respondent number across both schools	124	150	157	159

However, there were also cases of student dropouts. Some students got married during our 6-month tenure. Others dropped out to work in local gold mining operations in Kurutiyange. Student absenteeism and dropouts impacted the numbers of students participating in different data collection exercises and accounted for unsubmitted or lost questionnaires.

Student/Caregiver flock questionnaires and data

Out of the targeted 171 students, only 124 were present to fill the student questionnaire and only 150 students managed to fill the caregiver questionnaire by interviewing their parents/ caregivers. Besides being a learning activity for students, another purpose of the questionnaires was to obtain data about the household chicken flocks.

Using the collected data, household flocks were disaggregated by the numbers of roosters, hens and chicks. Proportions of roosters, hens, and chicks were, as expected, with households owning over twice as many hens as roosters, and about two chicks for each hen owned. Results from the students' initial questionnaires showed that female students had disproportionately fewer chickens than their male counterparts in both villages (Table 2). Students responded to data about flock sizes on 26/10/2022 through answering a questionnaire themselves, and also on 4/11/2022, when they interviewed their caregivers to collect more data about chicken farming (Table 3). Household flock sizes varied a lot over time between the two datasets, due to deaths, sales and misreporting of data. This came out strongly during the one-on-one sessions the team had with the students.

Table 2. Household ownership of chicken, totals according to student questionnaire

School	*Gender*	Roosters	Hens	Chicks	Total chicken
Kurutiyange	Female	75	168	274	517
	Male	163	337	562	1062
Kurutiyange Total		238	505	836	1579
Naora	Female	39	139	192	370
	Male	124	339	700	1163
Naora Total		163	478	892	1533
Grand Total		**401**	**983**	**1728**	**3112**

Table 3. Household ownership of chicken, totals according to caregivers' questionnaire

School	*Gender*	Roosters	Hens	Chicks	Total chicken
Kurutiyange	Female	63	166	229	458
	Male	125	414	535	1074
Kurutiyange Total		188	580	764	1532
Naora	Female	63	141	224	428
	Male	130	372	540	1042
Naora Total		193	513	764	1470
Grand Total		**381**	**1093**	**1528**	**3002**

Male students' households owned up to 115 chickens, whereas female student households owned a maximum of 46. Though most households had chicken, there were several students, especially in Kurutiyange who did not have chickens. This was mainly because they lived in rental properties around the village centre. Additionally, several homesteads experienced disease outbreaks before and during the study. This resulted in the loss of some flocks. Both factors disrupted learning for some students as they were not able to actively participate in the *T. vogelii* application experiment.

Care of Chicken

All the students who recorded having chicken in their household also recorded that they helped take care of the chicken. While mothers took a lead role in taking care of the chicken

in 63% of the households with chicken (Figure 2), 90% of students reported being involved in the feeding of chickens, which included giving food and water (Figure 3). A large proportion of boys in Naora (63%) were involved in the maintenance of the chickens which included cleaning and repairing the poultry house, chasing away predators (Table 4).

Figure 2. Primary carer for chicken in the students' households according to student questionnaire

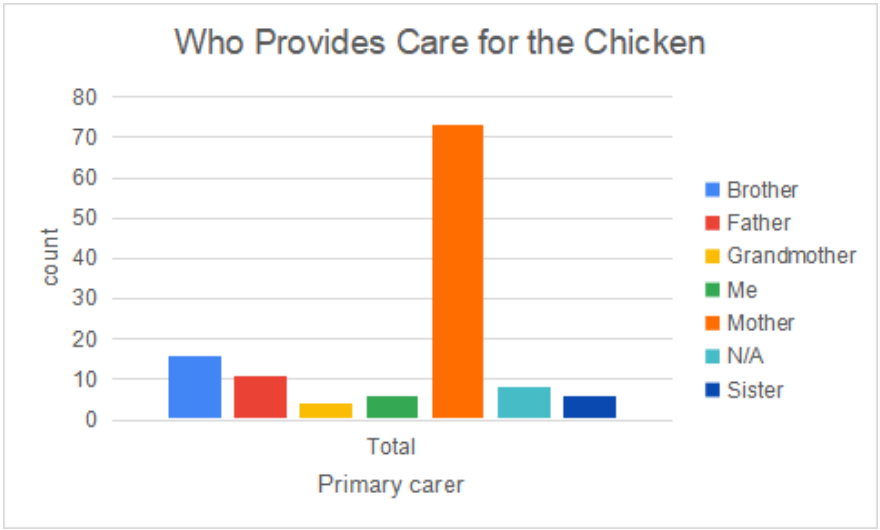


Figure 3. How students helped take care of chickens.

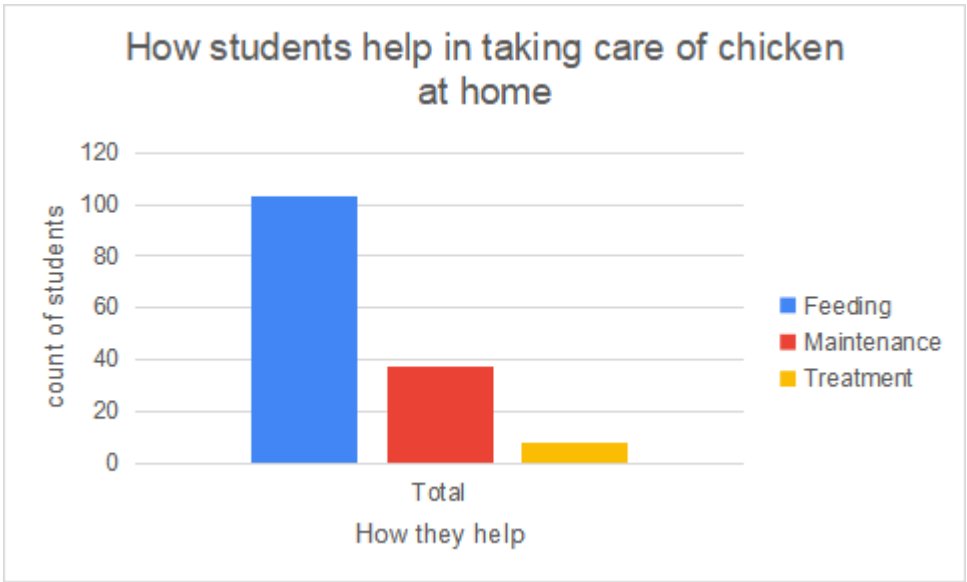


Table 4. Student involvement in the care of chicken, by school and gender

School	*Gender*	Help care	Feeding	Maintenance	Treatment
Kurutiyange	Female	27	26	5	0
	Male	34	31	4	4
Kurutiyange Total		61	57	9	4
Naora	Female	22	19	8	4
	Male	32	27	20	0
Naora Total		54	46	28	4
Grand Total		**115**	**103**	**37**	**8**

Reasons for keeping chickens.

Figure 4 indicates that most households kept chickens for sale, with the least common reason being to use chickens as gifts.

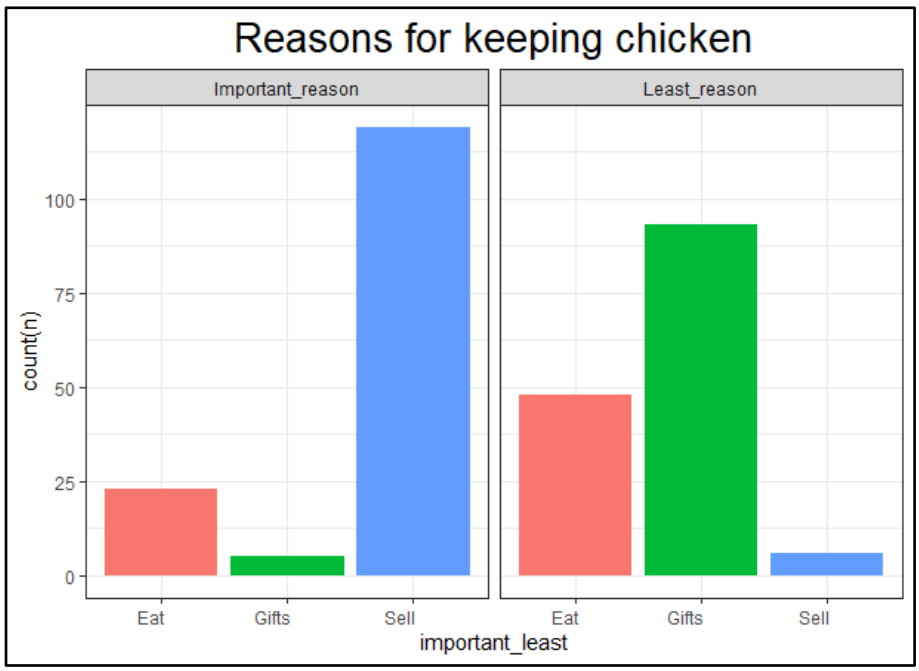


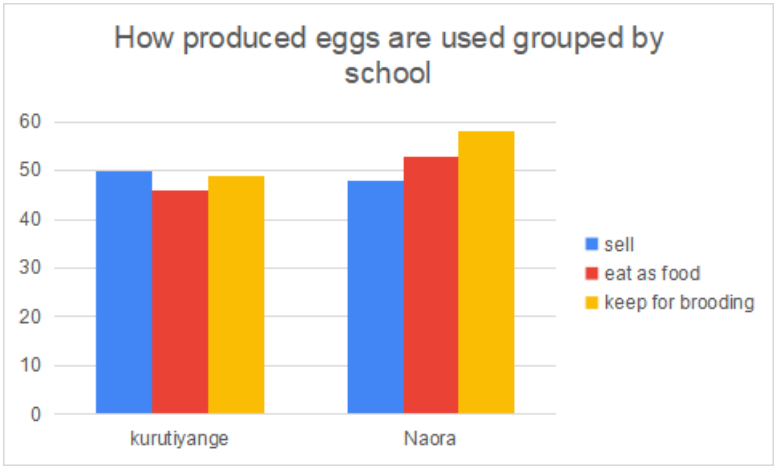
Figure 4. Reasons for keeping chickens according to caregiver questionnaire

How eggs are being used

From the caregiver questionnaire, it was evident that eggs are nearly equally valued for brooding, selling and for food (Figure 5). Minor uses for eggs included being given as gifts or used in barter trade for obtaining goods from local shops. Although the difference is

insignificant, more students from Kururtiyange reported that they sold their eggs and this can be attributed to the school being near a market centre as compared to Naora.

Figure 5. Reasons for egg production grouped by schools



According to the caregiver data more than half of household hatched chicks from reached maturity, while a small percentage of respondents reported a lower success rate, indicating that less than a quarter of hatched eggs resulted in mature chicks.

Factors contributing to loss of chickens.

The decrease in flock size in many households can be attributed to various factors according to the caregivers. Pests and diseases were the main contributors to the loss of chickens, followed by predators such as hawks, dogs, cats and eagles. This information tracks with the results of the Kenya AE Hub Baseline Chicken survey.

Student challenges with questionnaires

Students learned how to fill questionnaires and developed their interviewing skills by collecting data from their parents (Figure 6). This proved a challenging task for most of the students since it was their first time. Additionally, while interviewing their parents, some students encountered challenges which are also faced by data collectors, such as uncooperative respondents and false response. Since they did not know how to proceed after such a response, some students opted to fill in their questionnaires themselves. Through group discussions that were held in class after the interviews, the students were able to come up with ways such as involving their siblings in their school work.

" *My mother said that since she did not go to school she could not answer my questions. When I kept insisting, she became angry and chased me away. I decided to ask my elder brother who helped me with the answers*." Chacha, student at Naora Primary School



Figure 6. Student filling in a questionnaire

Insights from home visits

Home visits enabled the team to check student progress on their research activities. During the first phase, the visits brought to light issues with student participation: some students had already pre-filled the questionnaires, while others had not yet started by the time we visited.

Most entries were also incorrect, as some students had not understood or misinterpreted the questionnaire. For instance, while students were required to make entries of eggs laid on a daily basis, entries made were a mixture of cumulative eggs laid and daily eggs laid by the chickens. In other cases there were records of zero activities but upon follow up it was discovered that some occurrences like loss of chicken, brooding chickens hatched eggs etc. were not entered. This made the observation journals incomplete and inconsistent by the time they were collected from the students. Unfortunately, this made the data collected during the two month holiday unusable.

During home visits, the team observed interesting indirect outcomes of the study on students. While some students already had their own flock, 3 boys from Naora and 1 girl from Kurutyange primary schools grew their flock sizes. Parents had observed that these students had taken more interest and participated more in activities like providing food and clean water, maintaining hygiene by cleaning the sleeping area, and providing medicine to their chicken on time. Other students, mostly boys from Naora, had implemented what they learned by building chicken coops (Figure 7) and observing the health of their flocks more keenly, for earlier treatment of parasites.



Figure 7. Chicken coops built by students in their homesteads

Most parents valued the visit, leading to enhanced comprehension of the project and their role in facilitating their child's education (Figure 8). Chacha, a parent of one of the students in Naora, appreciated that the project helped him follow the progress of his son, since the student improved on attending school, as well as completing and seeking help doing assignments. Additionally in Phase 2 (after the first visit), some students planted a lot of *T. vogelii* and also engaged with other FRN members within their community to learn more about the use of *T. vogelii*. Parents were happy that their children were getting knowledge of using *T. vogelii* which they said would be a good method of addressing disease prevention and management in their homes.



Figure 8. Parent listening in while the student explains about the project during a home visit

Questions regarding personal benefits from the project, such as obtaining chickens or sponsorship, suggested that some parents may have misconstrued the project's purpose or held differing expectations. A parent from Naora primary school said:

"If my son does well in this project, what will he get? I would like it if he is gifted school uniforms and books since I cannot afford buying. "

Parent, Naora Primary School

Other parents from Naora and Kurutiyange also suggested that they get chickens at the end of the experiment. One parent said:

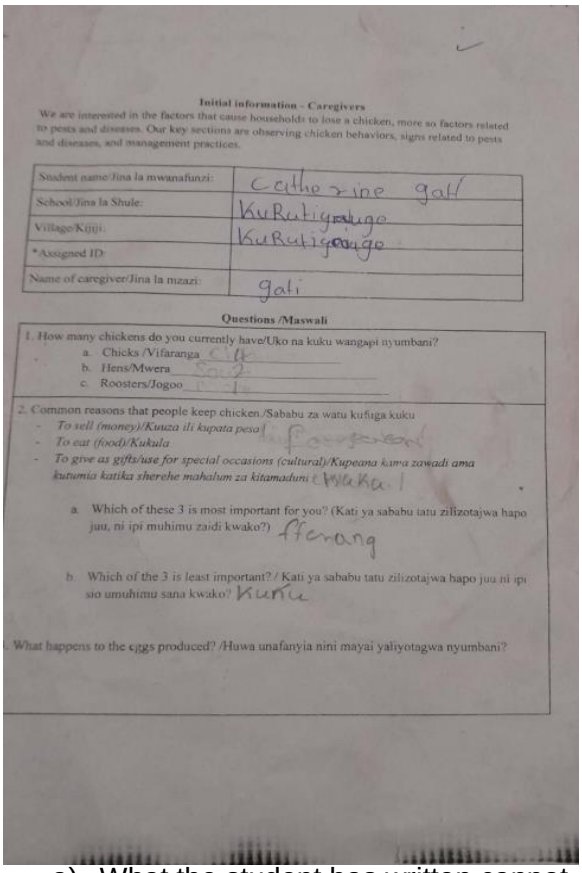
"I think it would be better if you give my child chicken so that he can start a business on his own. Otherwise what you have taught him will be useless if he cannot practise it after you have gone."

Team challenges with the questionnaires

Data collected from students' initial questionnaires was very difficult to interpret and required cleaning and corroboration. For instance, as shown in Figure 7a, some students did not know how to write their names properly, or provided nicknames rather than their official names. The challenge of poor and illegible handwriting slowed the data collection process. The students were introduced to reading and writing exercises, flashcards and speed tests at the beginning of the lessons and supervised composition writing to improve their skills. They were also encouraged to write down their answers on the board to build their confidence and presentation skills. However, this strained the limited time the team had to implement the project.

A small number of students deliberately provided false responses, including that they didn't have chicken at home, or by under-reporting and over-reporting their flock sizes. We found out from the parents and during some discussions with their teachers that the students had a habit of not doing their assignments because follow up was not usually done. These issues necessitated a review of the entries with the students to clarify before digitisation of the data by the research team. In the case of holiday journaling, the data obtained was never used because it had too many errors. In Figure 7b, the entries made by the student showed a lot of inconsistencies as the days went by. The questionnaire was untidy and also incomplete and this posed a big challenge when it came to making entries in the online forms.

Figure 7. The questionnaires showing how students made entries



Initial information – Caregivers

We are interested in the factors that cause households to lose a chicken, more so factors related to pests and diseases. Our key sections are observing chicken behaviors, signs related to pests and diseases, and management practices.

Student name/Jina la mwanafunzi:	Catherine gati
School/Jina la Shule:	KuRutigatugo
Village/Kijiji:	KuRutigatugo
*Assigned ID:	
Name of caregiver/Jina la mzazi:	gati

Questions /Maswali

1. How many chickens do you currently have/Uko na kuku wangapi nyumbani?
   a. Chicks /Vifaranga 6
   b. Hens/Mwera 2
   c. Roosters/Jogoo 1

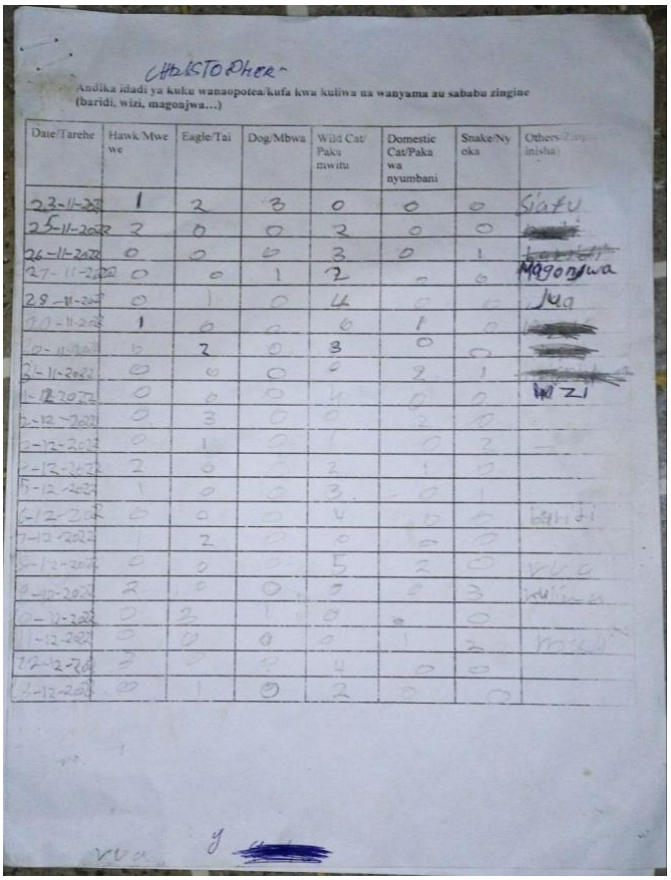
2. Common reasons that people keep chicken./Sababu za watu kufuga kuku
   - *To sell (money)/Kuuza ili kupata pesa* Food
   - *To eat (food)/Kukula*
   - *To give as gifts/use for special occasions (cultural)/Kupeana kama zawadi ama kutumia katika sherehe mahalum za kitamaduni* Waka

   a. Which of these 3 is most important for you? (Kati ya sababu tatu zilizotajwa hapo juu, ni ipi muhimu zaidi kwako?) Kurua

   b. Which of the 3 is least important? / Kati ya sababu tatu zilizotajwa hapo juu ni ipi sio umuhimu sana kwako? Kuua

What happens to the eggs produced? /Huwa unafanyia nini mayai yaliyotagwa nyumbani?

a) What the student has written cannot be understood



Andika idadi ya kuku wanaopotea/kufa kwa kuliwa na wanyama au sababu zingine (baridi, wizi, magonjwa...)

Date/Tarehe	Hawk/Mwewe	Eagle/Tai	Dog/Mbwa	Wild Cat/Paka mwitu	Domestic Cat/Paka wa nyumbani	Snake/Nyoka	Others/Zingine (nisha)
23-11-2022	1	2	3	0	0	0	Siafu
25-11-2022	2	0	0	2	0	0	[redacted]
26-11-2022	0	0	0	3	0	1	[redacted]
27-11-2022	0	0	1	2	0	0	Magonjwa
29-11-2022	0	1	0	4	0	0	Jua
30-11-2022	1	0	0	0	1	0	[redacted]
1-12-2022	0	2	0	3	0	0	[redacted]
2-12-2022	0	0	0	2	2	1	[redacted]
3-12-2022	0	0	0	4	0	0	Wizi
4-12-2022	0	3	0	0	2	0	
5-12-2022	0	1	0	1	0	2	
6-12-2022	2	0	0	2	1	0	
7-12-2022	1	0	0	3	0	1	
8-12-2022	0	0	0	4	0	0	Baridi
9-12-2022	0	2	0	0	0	0	
10-12-2022	0	0	0	5	2	0	Wizi
11-12-2022	2	0	0	1	0	3	Wizi
12-12-2022	0	2	1	0	0	0	
13-12-2022	0	0	0	0	1	2	Magonjwa
14-12-2022	2	1	0	4	0	0	
15-12-2022	0	1	3	2	0	0	

b) The work done is incomplete, incorrect, untidy and the student provided only one name.

The caregivers' questionnaire also had a different flock size compared to the previous exercise. During followed-up sessions with students, it was noted that the change in numbers was due to buying, selling, death or consumption of the chicken, which the students failed to record. Chicken which were slaughtered at home never made it to the records since some students did not consider it a loss in the flock unless it was a really huge number. It was also observed that some students living in the same homestead but with different households encountered challenges in recording the number of chickens they owned. Other instances included students providing false responses by guessing answers to the questionnaire, either because they did not interview their parents or because the parents refused to answer the questions.

Tephrosia vogelii experiment results

Students participated in a 6-week long experiment by making observations about ectoparasites management and behaviour of chicken which received tephrosia treatment. Data from the experiment was collected by questionnaire, and qualitative observational recordings completed in a daily journal.

For the student's experimental *Tephrosia vogelii* project, observations of pests before and after intervention were collected for two weeks and four weeks respectively. Being their first major experiment to do, the students took a while understanding how to make meaningful entries and the concept of consistency in data collection. Since the experiment was done at home, close supervision of the experiment was not possible and that meant the team had to be constantly reviewing and discussing the dos and don'ts with students. Ultimately, we were forced to conclude that the data from this experiment could not be confidently used to draw conclusions on the effectiveness *T. vogelii* in controlling ectoparasites.

During the second home visit, many students had said that their parents helped them in the application of *T. vogelii* and monitoring the chickens. However, there were cases where parents had stopped their children from giving the treatment, saying it was not effective or not trusted. This is mainly because the plant is known as fish poison and parents did not trust it to treat chicken. Other parents reported that they had given the treatment but their chicken was still dying and blamed it on the *T. vogelii*. On further inquiry, it was discovered that there was a disease outbreak in Naora village which was killing mainly chicks.

In the end, when comparing data collected from previous studies done by RFRN farmers and the data obtained from *T. vogelii* experiment done by the students, it was evident that students were not collecting better quality data than FRN members.

Conclusions

In conclusion, based on the challenges encountered, more effort, time and resources are needed if students are to become useful partners for RFRN data collection. However, we did not consider the project a failure. The student learning was evident based on the improved handwriting, more participation in home and school activities, improved attendance and feedback from both parents and teachers.

Students were able to gain skills that were implemented both at home and in their school work. The learning process was slow as many students had challenges with reading and writing and using technology. Additionally, most of them missed school and this interfered with their learning progress. Assignments given to students helped improve students' basic skills. For instance, students wrote compositions regularly and it appeared to help align their thought processes and improve their thinking and observation skills. With the improvement of handwriting, some students made better entries on the subsequent data entry activities.

Home visits also provided great insight into their home life and how it affected the students' learning. The economic activities differed greatly in the two villages, with Kurutiyange heavily involved in gold mining, while Naora had farming as the main activity. Observations made during the home visits suggested that some students did not receive the support of their parents in collecting information about their chicken. This was mainly in Kurutiyange, where there was limited interaction between students and parents, as the latter spent most of their time in the mining fields. Incorporating parents in the beginning of Phase 2 proved beneficial in implementation of the pilot study. This is because most parents who had little knowledge about the project were reluctant to support their children in learning in the first phase of the pilot study. The visits further improved parents' understanding of the importance of being involved in their students' education.

Appendices

A. Data collection tools

Initial information: completed by students

Initial information - Students

Student Name:	
School:	
Village Name:	
* Student ID *(Get information from the class register):*	
Parent/Guardian Name:	
*Is parent/guardian FRN member *(Get information from Chacha):*	
Do you have chickens at home? (*Yes/No*)	
Who normally takes care of the chickens?	
Do you help take care of the chickens? (*Yes/No*) How?	
Approximately how many chickens do you have? (Give a number) Roosters = Hens = Chicks =	

B. Initial information: students interviewed their caregivers

Initial information - Caregivers

We are interested in the factors that cause households to lose a chicken, more so factors related to pests and diseases. Our key sections are observing chicken behaviours, signs related to pests and diseases, and management practices.

Student name/Jina la mwanafunzi:	
School/Jina la Shule:	
Village/Kijiji:	
*Assigned ID:	
Name of caregiver/Jina la mzazi:	

Questions /Maswali

1. How many chickens do you currently have/Uko na kuku wangapi nyumbani?

    a. Chicks /Vifaranga_____
    b. Hens/Mwera_____
    c. Roosters/Jogoo_____

2. Common reasons that people keep chicken./Sababu za watu kufuga kuku

- *To sell (money)/Kuuza ili kupata pesa*
- *To eat (food)/Kukula*
- *To give as gifts/use for special occasions (cultural)/Kupeana kama zawadi ama kutumia katika sherehe mahalum za kitamaduni*

    a. Which of these 3 is most important for you? (Kati ya sababu tatu zilizotajwa hapo juu, ni ipi muhimu zaidi kwako?)

b. Which of the 3 is least important? / Kati ya sababu tatu zilizotajwa hapo juu ni ipi sio umuhimu sana kwako?

3. What happens to the eggs produced? /Huwa unafanyia nini mayai yaliyotagwa nyumbani?

4. For every 10 eggs your hens produce, how many would you typically…/ Kwa kila mayai kumi kuku hutaga, ni ngapi…

- Incubate (leave for hen to hatch into chicks)/Humwekea kuku kulalia ili aangue vifaranga_____
- Sell/Huuza_____
- Eat with the family/Huliwa na familia_____

5. How many of your chicks typically reach maturity? /Kwa kawaida, ni vifaranga wangapi ambao hukuwa wakubwa

- less than a quarter,/chini ya robo
- a quarter to a half, /kati ya robo na nusu
- more than half /zaidi ya nusu

6. What are the most important factors that cause you to lose birds from your flock? (rank from most to least important; if there are specific diseases that are very important, give the local name if you know it) / Sababu gani huchangia kupoteza kuku wako? Taja ukianzia sababu kuu na ukimaliza na sababu nyingine. Iwapo kuna vifo vya kuku ambavyo vinasababishwa na magonjwa unazojua yataje kwa jina)

7. What time of year (season) do you lose the most birds? /Ni wakati/msimu/miezi ipi wa mwaka huwa unapoteza kuku wengi sana?

C. Student daily log: completed by students during their holiday

https://docs.google.com/document/d/1h1pm2eqECHLbxyn2WsRBEHmSfiNfV6rmDa01NKkN-xM/edit?usp=sharing

Daily log: number of birds lost to specific predators and other factors (cold weather, theft, disease, etc)

Date	Hawk	Eagle	Dog	Wild Cat	Domestic Cat	Snake	Disease	Other eg theft, cold, etc (specify)	Total number of birds left

Daily log: number and source of new birds in the flock and record of eggs

Date	Chicks hatched	Gift	Purchased	Number of eggs laid
4 Nov	5 chicks	2 Hens	0	4

Daily log: number of birds in the flock, losses and gains, and record of eggs

Date	Total Chicks	Total Hens	Total Roosters	Animals lost (reason)	Animals gained (reason)	Eggs laid

D. Kiswahili versions of daily logs

Daily observations checklist:

https://docs.google.com/document/d/1q-tkwMdMZtZ9pPFEyGNN4JZG5YCoM-OS0e2x45JV72s/edit?usp=sharing

Ishara za kimwili: Za kawaida na zisizo za kawaida

Ishara za kimwili	**Za kawaida** **(all of the following)**	**Zisizo za kawaida** **(any of the following)**
Appearance of the comb Muonekano wa undu/nyundu	Rangi nyekundu zaidi haina dosari	Undu umejaa viroboto Rangi nyekundu lakini ni nyepesi Undu una sehemu nyingine ambayo ni nyekundu Sugu
Macho	Macho ni safi Ukubwa wa macho ni kawaida	Macho yanatoka machozi/maji Macho kuvimba
Mapua	Imekauka	Maji au kamasi yanatoka kwenye mapua

Mabawa	Zinangara Unruffled and smooth Area around vent is clean	Kuku anajikuna mara nyingi na kuchuna mabawa akitumia kidona chake Mahali ambapo kuku hotolea kinyesi ni nyeusi na/au imevimba
Miguu	Safi na laini	Yana magamba
Mayai	Safi	Mayai yana madoadoa nyekundu au nyeusi Umbo lisilo la kawaida

E. <u>Daily journaling guide:</u>

https://docs.google.com/document/d/1ffuBnQkAZC5EADOYLbOrQMHIn7Nr61U-j1wB7QHJgdk/edit?usp=sharing

Orodha ya tabia za kuku za kawaida na zisizo za kawaida.
Mwanafunzi kurekodi uchunguzi wowote wa tabia ya kuku isiyo ya kawaida katika daftari yake.

Tabia ya kuzingatia	**Kawaida/normal** **(yote yafuatayo)**	**Isiyo ya kawaida/Abnormal** **(yoyote ya yafuatayo)**
Appetite/Hamu ya kula	• Kuku ana hamu ya chakula, anakula vizuri	• Hana hamu ya chakula na hali vizuri
Posture/Mkao	• Alert and on guard/ Anatahadhari	• Ameinama • Ameweka kichwa karibu na mwili • Amelala chini • Mabawa kushuka

Relationship with other chickens/Mahusiano na kuku wengine	• Anazunguka kwa juhudi na kundi	• Anajitenganisha na kundi. • Hana nguvu. • Anaonekana mwenye huzuni (amefunga macho, manyoya hayajajipaga, hatembei tembei).
Movement/ Harakati	• Anatembea na kukimbia akigwara mchanga..	• Harakati za ndege hazina usawa. • Ndege anaweza kujikwaa. • Hataki kutembea mbali.
Breathing/Kupumua	• Kawaida (Hakuna cha kutambua)	• Kupumua, kupiga chafya au kukohoa
Dust bathing/Kuoga vumbi	• Kawaida (Hakuna cha kutambua)	• Mara nyingi wakati wa mchana
Laying eggs/Kutaga mayai	• Anataga mayai mara kwa mara(mayai ya kawaida) mahali alipotengenezewa kioto.	• Kuku hutoroka mahali alipotengenezewa kutaga mayai (kioto)
Brooding/Kulalaia mayai	• Kuku analalia mayai mpaka wakati anapoangua vifaranga.	• Kuku anatokatoka mahali pa kulalia mayai(anatoroka kwenye kioto).
Chicken behaviour when they go to sleep or wake up / Tabia za kuku wakati anapoenda kulala au kuamka	• Ana hamu kwenda katika eneo la kulala na kukaa kule.	• Kusitasita kwenda kwenye eneo la kulala.

Andika idadi ya kuku wanaopotea/kufa kwa kuliwa na wanyama au sababu zingine (baridi, wizi, magonjwa…)

Date/ Tarehe	Hawk /Mwewe	Eagle/ Tai	Dog /Mbwa	Wild Cat/ Paka mwitu	Domestic Cat/ Paka wa nyumbani	Snake/ Nyoka	Others/ Zingine (Bainisha)

Andika idadi ya kuku wageni kwenye kundi na rekodi ya mayai yanayotagwa kila siku.

Date/ Tarehe	Hatched chicks/ Vifaranga walioanguliwa	Given as a gift/ Kupewa kama zawadi	Bought/Kuku Kununuliwa	Number of eggs laid/ Idadi ya mayai yaliyotagwa

F. Observation sheet (Used in phase 2):

https://docs.google.com/document/d/1ErCZ4qK-oJs2pZsMoCGf3ZYlKWotj-menXy2obn02sw/edit?usp=sharing

Students' Observation Sheet: English

Name: _____

School: _____

Date: _____

1. How many chickens do you have? Roosters = …. Hens =.... Chicks = ...
2. How many eggs have been laid? *(between the last observation and today)* ....
3. Do you have any hens brooding? (Yes / No)
4. How many eggs are in the brooding nest?
5. How many eggs have hatched? *(between the last observation and today)* ....

Observed parasites: Have you observed the following parasites on the tagged chickens?

Parasites	Observed parasite		How many parasites have you observed on the tagged chickens?		Of the tagged chickens, how many did you observe having the parasite?
	Yes	No			
Fleas	●	●	Fleas =		
Ticks	●	●	Ticks =		
Mites	●	●	● Few	● Many	
Lice	●	●	● Few	● Many	

Physical signs on the tagged chickens: Have you observed the following physical signs of parasite infestation on the tagged chickens?

Physical Signs	Observed Physical signs		Students' Observation
	Yes	No	

Comb *i.e.Fleas attached to the comb, Pale colour, Patchy red, Scabs or warts on comb*	●	●	
Eyes i.e. *Discharge of fluid from eyes, Eyes swollen*	●	●	
Feathers i.e. *Frequently scratching self and excessive preening of feathers*	●	●	
Under the wings *i.e mites and eggs at the base of the feathers[*	●	●	
Vent i.e. *Area around the vent is blackened or inflamed*	●	●	
Legs i.e *Scaly legs*	●	●	
Eggs i.e *Red or black specks on the eggs or reduced egg laying*	●	●	

G. Kiswahili observation sheet

Karatasi ya Uchunguzi: Kiswahili

Jina: _____

Shule: _____

Tarehe: _____

Una kuku wangapi? Jogoo = …. Kuku =.... Vifaranga = ...

Vimelea vinavyozingatiwa: Je, umeona vimelea vifuatavyo kwenye kuku waliotambulishwa?

Vimelea	Vimelea vinavyozingatiwa		Umeona vimelea wangapi kwenye kuku waliotambulishwa?		Kati ya kuku waliotambulishwa, ni wangapi uliowaona wakiwa na vimelea hivyo?
	Ndiyo	Hapana			
Viroboto	●	●	● Baadhi	● Wengi	
Utitiri	●	●	● Baadhi	● Wengi	
Kupe	●	●	● Baadhi	● Wengi	
Chawa	●	●	● Baadhi	● Wengi	

Ishara za kimwili kwenye kuku waliotambulishwa: Je, umeona dalili zifuatazo za uvamizi wa vimelea kwenye kuku waliotambulishwa?

Ishara za Kimwili	Ishara za Kimwili zinizozingatiwa		Uchunguzi wa Wanafunzi
	Ndiyo	Hapana	

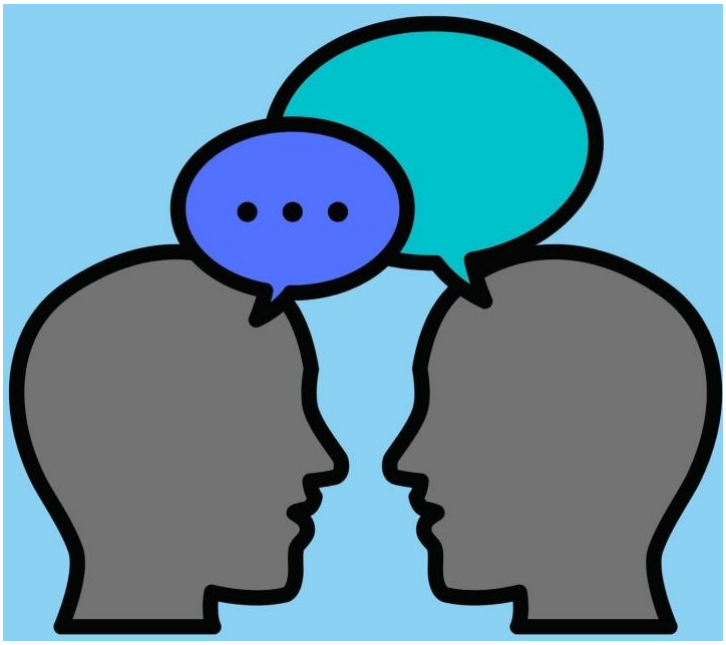
Sega ya kuku *i.e.Viroboto vinavyokatalia kwenye sega,Rangi iliyofifia,mabaka nyekundu kwenye sega, ,Vipele au uvimbe kwenye sega*	●	●	
Macho i.e.*Kutokwa na maji kwa macho,Macho kuvimba*	●	●	
Manyoya ya kukui.e.*Kujikuna mara kwa mara na kusafisha manyoya kupita kiasi*	●	●	
Tako la kuku i.e.*Eneo karibu na tako ni nyeusi au limevimba*	●	●	
Miguu i.e *magamba kwenye miguu*	●	●	
Mayai *Madoa nyekundu au nyeusi kwenye yai*	●	●	

H. General observation:

https://docs.google.com/spreadsheets/d/1cJez_YV3rhDvtzj9TK_0bGUljwU_jPebuZegC8keZsA/edit?usp=sharing

A. Teaching materials/lesson activities

COMMUNICATION



UNIT 1:

COMMUNICATION

Description: Communication is the act of transferring information from one person to another, whether vocally, visually, or non-verbally. It is a core competency skill promoted by the Kenyan Competency-Based Curriculum (CBC). All the developed lessons in this project aim at promoting skills that cut across different subjects. The activities within the lessons have a direct or indirect link to the Competency Based Curriculum. Additionally, sub-strand 2.1 ***(Practices in rearing small domestic animals)*** in the Grade six agriculture subject, is very vital to the project.

In this unit, learners will develop communication skills through engagement in the charade game and the telephone game. The unit can be linked to the CBC learning strands and substrands listed below;

- Agriculture, Strand 2.0 *(Domestic animals)*, Sub strand 2.1 *(Practices in rearing small domestic animals).*
- English, Strand 1.1 *(Listening and Speaking)*, Sub strand 1.1.1 & 2.1.1 *(Pronunciation and Vocabulary: Listening comprehension).*
- English, Strand 4.2 *(Reading)*, Sub strand 4.2.1 *(Intensive Reading: visuals).*
- English, Strand 9.1 *(Listening and Speaking)*, Sub strand 9.1.1 *(Pronunciation and Vocabulary: Speaking Fluency).*
- English, Strand 10.1 *(Listening and Speaking)*, Sub strand 10.1.1 *(Pronunciation and Vocabulary: Interactive Listening).*
- Indigenous Languages, Strand 1.0 *(Listening)*, Sub strand 1.1 *(Attentive listening).*
- Indigenous Languages, Strand 1.0 *(Listening)*, Sub strand 1.4 *(Extensive listening).*

- Indigenous Languages, Strand 2.0 *(Speaking)*, Sub strand 2.4 *(Public speaking).*

As it is, the communication discipline focuses on how people use messages *(vocally, visually, or non-verbally)* to generate meanings within and across various contexts, cultures, channels, and media. Therefore, the skills gained by the learners are essential to the project in that they will be able to observe chicken behaviour, physical manifestation, and poultry management practices at home related to pests and diseases, collect information about the behaviour/management practices, and effectively communicate the behaviour/management practices.

Objective: Learners to understand the importance of keenly observing behaviour, collecting information about the behaviour, and effectively communicating the behaviour.

Skills acquired: observation, communication, collaboration, and listening skills.

Estimated time: 1 hour 20 minutes over two days.

Key question: Why is it important to communicate effectively?

Materials: Notebook, pen/pencil, chalk

Activities: This unit has been divided into the following lessons;

1. The Telephone Game
2. The Charade Game

LESSON 1: The Telephone Game	**40 Minutes**

Learning outcomes for this lesson:

- Learners will be able to keenly listen to the information given to them and pass it on to the next learner.

Learning experience:

(20 minutes)

a) Learners demonstrate attention by listening to others and communicating exactly the same information to their peers. Ideas for the game can include;
   - Clever cats can count / Paka mwerevu anaweza hesabu
   - Ready, set, go!
   - Find four funny fish / tafuta samaki wanne wanaochekesha
   - This little chick ate corn today / Kifaranga mdogo amekula mahindi leo
   - Cluck cluck, red hen / Cluck cluck, kuku mwekundu
   - Where is the fat chicken? / Kuku mnene yuko wapi?
   - Spoiled chickens lay better eggs / Kuku anayetunzwa vizuri hutaga mayai mengi
   - You're a good egg / wewe ni yai nzuri
   - Don't ruffle my feathers / usishike manyoya yangu

- A fat chicken is a healthy chicken / Kuku mnono ana afya nzuri

Playing the telephone game:

- Organise the learners into groups and assign them group names.
- In the groups, the person starting the game thinks of a word or phrase and whispers it into the next player's ear only once, with no repeats allowed.
- That listener tries to correctly repeat that same word or phrase into the next player's ear. The last person in the line or at the end of the circle repeats the phrase or word aloud.
- Allow a moment for giggles if the message is "broken" or changed. The player who started announces the correct word or phrase.

(10 minutes)

b) The facilitator to guide the group members in discussing why information passed from the first person in (a) above is the same or different from that passed to the last person. Facilitator should expect learners to: *blame each other if the phrases are different from the expected, claim not to hear the phrases*

Guide Question: Why do you think the phrases passed from the first person to the last person are the same or different?

(10 minutes)

c) Learners to justify why it's important to keenly listen and accurately communicate information. Facilitator to let a few of the learners to list their ideas and share them with the rest of the class.

Guide question:

- What happens when people do not listen attentively?
- What are the consequences of not communicating effectively?

Reflection Question:

1. Did you learn anything today? What did you learn?

LESSON 2: The Charade Game	**40 Minutes**

Learning outcomes for this lesson:

- Learners will be able to demonstrate actions without using words.
- Learners will be able to keenly observe non-verbal communication from their peers and interpret its meaning.

Learning experience:

(25 minutes)

a) Learners demonstrate actions without using words or draw and allow their peers to interpret the meaning of those actions/drawings. This should include chicken

behaviours, signs, and poultry management practices they do observe at home e.g. *roosting, dust bathing, flapping wings, brooding, foraging, shivering, etc.* The table below provides a few examples.

Jogoo	eating	running	slashing	writing	monkey
Kuku kulalia mayai	spraying	dancing	Milking	teacher	Duck (Bata)
flying	Weeding	drinking	chicken	choir master	football
Roosting (jogoo kuwika)	kutetemeka	sleeping	sweeping	frog	brushing teeth

Playing the charade game:

1. Organise the learners into groups of equal numbers and assign them group names.
2. Choose a group to go first.
3. The group that goes first chooses one group member to go first.
4. Choose the word or phrase for the team member to act out without words.
5. The selected member acts out the word for the rest of the group to guess the word or phrase within one minute. Talking is not allowed.
6. If the group is able to guess the word or phrase, they get a point. If they don't guess correctly, they don't get a point.
7. The other groups repeat this process.
8. Keep playing until all group members have gone a few times. If there are too many groups, multiple groups can play at the same time, with at least another group 'supervising'.
9. The team with the most points at the end of the game wins.

(15 minutes)

b) In groups, learners discuss and note down what they have learned from the game, i.e., why they won or lost the game and present it to the class.

What I learned from the game?	Why we won or lost the game?

Reflection Question:

1. Did you find the game interesting? How?
2. Is there anything that you have learnt from this game?
3. How will you make the game more interesting?

N/B: Facilitator makes sure learners understand what he/she is trying to illustrate using the two lessons and how it would be useful when handling the chicken.

LESSON 3: Doing Presentations	**40 Minutes**

Reflection questions for this Unit:

1. Which activity helped you to learn more? Why
2. What did you learn in this lesson?

Baseline information assessment

Way forward…

Learners to observe chicken behaviours/poultry management practices at home and act out the behaviours/practices with their friends/family using the charade game as a guide.

DATA RECORDING



"Write It Down."

DATA RECORDING

Description: The skill of recording data involves the documentation of observations in a variety of forms in order to preserve them for later use. The recorded observations are then analysed, visualised, and interpreted so as to get useful information. The skill promotes the development of a wide range of written, oral, and visual skills involved in documenting and organising information.

All the developed lessons in this project aim at promoting skills that cut across different subjects. The activities within the lessons have a direct or indirect link to the Competency-Based Curriculum (CBC). Additionally, sub-strand 2.1 ***(Practices in rearing small domestic animals)*** in the Grade six agriculture subject, is very key to the project.

Recording observations is important for the learners to keep track of the changes around them, thus helping in getting new meaning and significance. In addition, when a learner keeps records, they get to know what to look for and ensure an accurate description of the observations made.

Skills acquired in this unit cut across different subjects and the lessons can be linked to the CBC learning strands and substrands listed below;

Grade 5

- Agriculture, Strand 2.0 *(Domestic animals)*, Sub-strand 2.1 *(Practices in rearing small domestic animals).*
- Mathematics: Strand: 4.0 *(Data Handling)* ,Sub-strand: 4.1 *(Data Representation)*
- English: Strand 1.4 *(Writing)*, Sub-strand 1.4.1 *(Functional Writing)*, 1.4.1.1 *(Filling forms)*
- English: Strand 4.2 *(Reading)*, Sub strand 4.2.1 *(Intensive Reading: visuals)*.

Grade 6

- Mathematics: Strand 2.0 *(Measurement)*, Sub-strand 2.1 *(Length)*
- Mathematics, Strand 4.0 *(Data handling)*, Sub-strand 4.2.1 *(Intensive Reading: visuals)*.

- English: Strand 1.4 *(Writing),* Sub-strand 1.4.1 *(Guided Writing)*

This unit builds up on the unit 2 lesson activities where learners acquire skills in making observations and select the chicken behaviours and poultry management practices they will be keen on observing. They will be introduced to the project's daily data tool and approaches to organising their observations in a clear and accurate way.

Objective: Learners to understand the importance of recording observations, how to use the project's daily data tool, and improving the quality of recorded observations.

Skills acquired: Collaboration, creative thinking, problem-solving, critical thinking, recording data, and documentation.

Key question: Why is it important to accurately and correctly record observations?

Estimated time: 2 hours over 3 days

Materials: notebook, pen, projector/printed photographs, chalk, blackboard

Activities: This unit has been divided into three lessons as follows;

1. Recording observations
2. Improving observation quality.
3. Daily data tool

LESSON 1: Recording observations — 40 minutes

Learning outcomes for this lesson:

- Learners will be able to explain why it's important to record observations.
- Learners will be able to recall chicken behaviours and poultry management practices to be observed and recorded.

Learning experience:

(5 minutes)

a) Ask learners to state instances where they have had to record data. In groups, learners share their experiences about what they collected and how they went about collecting that data.

Instance	How they collected

b) A few learners narrate the experience they had while brainstorming with their friends and families on the importance of recording chicken behaviours and poultry management practices.

(15 minutes)

c) In groups, learners discuss why we record information and select one member of the group to present to the rest of the class. Some of the expected responses may include: We keep records because:

"We do not want to forget information"
"We want to be accurate about some facts"
"Records may help us ask good questions"
"Explains some facts"
"To summarise some information"
"To share some information"

Guide question:

- Why is it important to record information?

(10 minutes)

d) In groups, learners identify different chicken behaviours and poultry management practices that they would be interested in observing and recording. Learners to identify and list the type of pests that they have ever observed in their chicken at home. They can give local names of the pests. The learners can also identify the signs and symptoms they observed in chicken that they think are associated with these pests. They can display the information in tables as shown below:

Pest(Local name)	Physical signs	

(10 minutes)

e) Learners make dummy recordings of the behaviours of their interest mentioned in (c) above in a table on their notebooks. The dummy data will be used in lesson 2.

Reflection question

1. What are some of the skills you have learnt today? How could you apply these skills in your daily life?
2. What do you find interesting about taking notes?
3. Do you think you are in a better position to help your parents at home take care of chicken? How?

LESSON 2: Improving data quality. 40 minutes

Learning outcomes for this lesson:

- Learners will be able to identify problems that could reduce the quality of their data records.
- Learners will be able to suggest possible ways of improving the quality of the records.
- Learners will be able to properly organise data records.

Learning experience:

(15 minutes)

a) In groups, learners keenly look at the displayed data *(sample below)*, identify potential errors in the data, and collaboratively correct the mistakes. Potential answers may include: *wrong entry on name in rows 1, 2, and 4, a wrong entry on grade in rows 1, 2, 4, a wrong entry on age in rows 1, etc.*

	Name	**Grade**	**School**	**Age**	**Height(cm)**	**Weight(kg)**
1	*Grade 6 Antony*	*Antony*	*Eleven*	*Nno*		
2	*Eleven*	*primary*	*7*	*eleven*	*don't know*	
3	*Joseph Chacha*	*six*	*St. Joseph's*	*12*	*140*	*50*
4	*46*	*for*	*Anita Lucy*	*10*	*130*	*primary*
5	*Maria Mburu*	*6*	*primary*	*13*	*cm*	*1500*

(15 minutes)

b) Each group organises and presents the dummy data they recorded in lesson 1 and present to the rest of the class for their peers to identify any errors in the data.

(10 minutes)

c) As a class, learners orally state potential problems that could prevent them from recording quality data on the chicken behaviours and poultry management practices. This can include poor storage of the records, *uncooperative parents/guardians, lack of writing materials, absence of chickens to observe, limited time to observe, unfamiliarity with English names, etc.*

Reflection Question

1. Is there anything new that you have learnt today? Get the students to write down their answers and stick them on the wall
2. What are the challenges you encountered during this lesson?
3. How did working in groups help you ?
4.

LESSON 3: Daily data tool **40 minutes**

Learning outcomes for this lesson:

- Learners to be introduced to the daily data tool which they will use to record observations on chicken behaviours and poultry management practices.

Learning experience:

(15 minutes)

a) Learners familiarise themselves with the daily observation tool i.e. what it looks like and how they will use it to record chicken behaviours and poultry management practices.

(10 minutes)

b) In groups, learners fill in dummy answers in the daily data tools and organise the dummy responses in a table.

(10 minutes)

c) The groups present their organised responses in a format of their choice and mention why they chose the format.

(5 minutes)

d) As a class, learners brainstorm on the importance of organising observations the way they did in (c) above.

Reflection question

1. Have you gained any skills today? Name them.

2. How do you plan to use the skills gained in this lesson to help your parents take care of chicken at home?
3. Do you think you are in a better position to help your parents at home take care of chicken? How?

Way forward…

As an assignment, learners use the daily data tool to observe and record chicken behaviours and poultry management practices at home. The recorded observations will be used in unit 4.



UNDERTAKING OBSERVATIONS

Description: Undertaking observations involves watching behaviours and events or noting the physical characteristics of something or a place in its natural setting. In this unit, learners will recall previous observations they have made on chicken behaviour; describe the behaviour, and try to connect the behaviour to the general poultry management practices at home.

All the developed lessons in this project aim at promoting skills that cut across different subjects. The activities within the lessons have a direct or indirect link to the Competency-Based Curriculum. Additionally, sub-strand 2.1 ***(Practices in rearing small domestic animals)*** in the Grade six agriculture subject, is very key to the project.

In this unit, learners will develop observation skills through pictorial and outdoor observation activities. The unit can be linked to the CBC learning strands and substrands listed below;

- Agriculture, Strand 2.0 *(Domestic animals)*, Sub strand 2.1 *(Practices in rearing small domestic animals).*
- English, strand 6.4 *(Writing)*, sub-strand 6.4.1 *(pictorial composition)*
- English, Strand 4.2 *(Reading)*, Sub-strand 4.2.1 *(Intensive Reading: visuals).*
- English, Strand 10.1 *(Listening and Speaking)*, Sub strand 10.1.1 *(Pronunciation and Vocabulary: Interactive Listening)*..
- Indigenous Languages, Strand 2.0 *(Speaking)*, Sub strand 2.4 *(Public speaking).*

This unit is important for the learners to observe chicken behaviour and poultry management practices at home, collect information about the behaviour/management practices, and effectively communicate the behaviour/management practices.

Objective: Learners are to be equipped with the knowledge and skills to make observations on behaviours and events in their natural setting and note the characteristics.

Skills acquired: Communication, collaboration, creativity, imagination, critical thinking.

Key question: How would you describe your chickens based on their behaviours?

Estimated time: 3 hours over four days.

Materials: notebook, pen, printed pictures (or projector).

Activities: This unit has been divided into four lessons as follows;

1. Learners' previous observations and knowledge of chickens
2. Pictorial observation
3. Outdoor observation (visiting a farm)
4. Introducing the project

LESSON 1: Learners' previous observations and knowledge of chickens (40 minutes)

Learning outcomes for this lesson:

- Learners will be able to describe behaviours in chickens that they have previously observed, either at home or elsewhere.
- Learners should be able to explain their understanding of previously observed chicken behaviours.
- Learners will be able to elaborate on their routine poultry management practices at home.

Learning experience:

(10 minutes)

a) As a class, learners list and discuss behaviours they have previously observed in chickens. Some of the notable behaviours may include *sunbathing, brooding, laying eggs, pecking, bullying, feeding, drinking water, roosting, and sounds (danger, crowing, greeting, warning, brooding, feeding*).

Guide questions:

- Tell us something about your chicken?
- What interesting things have you ever observed your chickens doing?

(15 minutes)

b) Ask the learners to think about why the chicken behaves in some of the ways they mentioned in (a) above. This can be done in groups and then allow a few groups to present to the rest of the class. At the end of each group's presentation, students from other groups will be allowed to comment/ask questions about the presentation.

Guide question:

- Why do you think chickens behave that way?
- What do you do when they display such behaviours?

Behaviour	What makes chickens show this behaviour?	Do you think that's a good behaviour (why)

(15 minutes)

c) Learners to mention and explain routine poultry management practices they have observed at home. This can be done in groups and then allow a few groups to present to the rest of the class. At the end of each group's presentation, students from other groups will be allowed to comment/ask questions about the presentation. Some of the management practices may include *cleaning the poultry house, feeding, and providing water*.

Guide questions:

- How do you always help your parents/guardians with managing the chickens? And why do you always do it?

Management practice	Why is it done?

Reflection Question:

1. What have you learnt from this lesson?
2. What did you find interesting about this lesson? Get the learner to mention/note down one thing they have learned from this activity
3. How has this lesson helped you understand what your parents do to take care of chicken at home?

LESSON 2: Pictorial Observation **(40 minutes)**

Learning outcomes for this lesson:

- Learners will be able to carefully observe and describe chicken behaviour in the provided images.
- Learners will be able to write an accurate descriptive excerpt of the images, with their thoughts organised logically, coherently, and clearly.
- Learners will be able to collaborate and improve on their written excerpts after examining each other's work.

Learning experience:

(5 minutes)

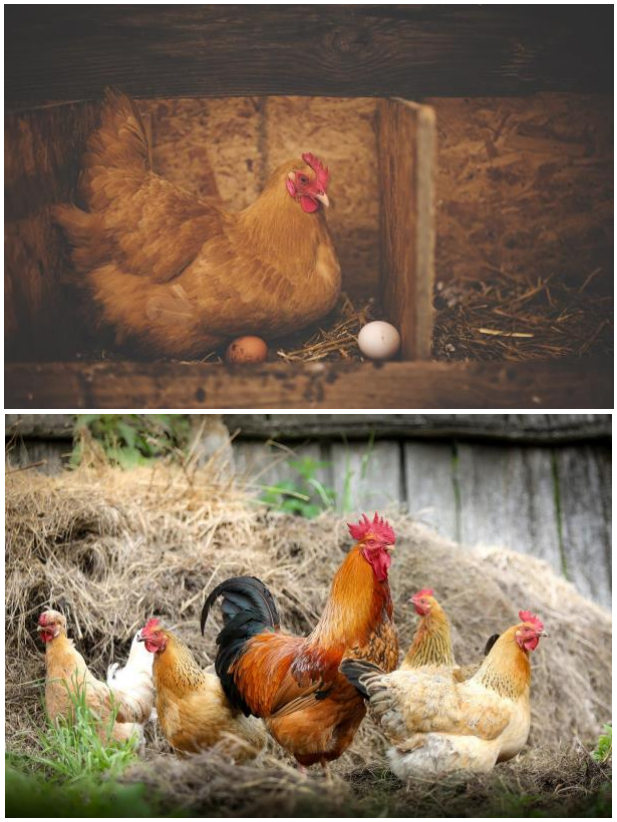
a) As a class, learners view Picture 1 *(below)* and briefly write down anything they observe(what is going on in the picture).



Picture 1: (source: Pixabay.com)

(20 minutes)

b) In groups, learners view the pictures below and explicitly note down descriptions of what they see in the pictures, then select one of the group members to present their descriptions to the rest of the class.



Picture_2 (source: Pixabay.com)

(10 minutes)

c) Learners comment on each group's presentation.

(5 minutes)

d) The facilitator choses a student to take note of the ideas being presented by the learners and then highlight them after all group presentations have been commented on by the learners.

Reflection Questions:

1. What have you learnt from this activity?
2. Can you relate any of the observations to what happens at home? Please give some examples
3. Which activity helped you learn the most about chicken? Why?

LESSON 3: Introducing the project (40 minutes)

Learning outcomes for this lesson:

- Learners will be briefed on the project objectives..
- Learners will be informed of the major chicken behaviours and poultry management practices to be observed at home.

Learning experience:

(30 minutes)

a) In detail, the facilitator gives an overview of the project and the role of the learners in the project. The facilitator should allow the students to seek clarification about the project.

(10 minutes)

b) As a class, learners have a brainstorming session to discuss chicken behaviours and poultry management practices that they will be keen on observing at home. Some of the behaviours may include *sunbathing, brooding, laying eggs, pecking, bullying, feeding, drinking water, roosting, and sounds (danger, crowing, greeting, warning, brooding, feeding*).

c) The learners list behaviours of a healthy chicken they have ever observed. Ask the learners how they know that a chicken is not well. Observations made on behaviour displayed by sick chicken may include the following: Observations on physical manifestations of pests and diseases may include: *swelling around the eyes, scaly feet, swollen feet, fleas around the eyes*
The learners create a table displaying the behaviours of healthy chicken and sick chicken.

Chicken behaviour

Healthy chicken	Sick chicken

Guide question:

- Which chicken/poultry behaviour would you record at home?

Reflection question:

1. What have you learnt about this project?
2. Why do you think this project is important to you as a student?

LESSON 4: Outdoor observation (visiting a farm) (60 minutes)

Learning outcomes for this lesson:

- Learners will be able to carefully observe and describe chicken behaviour and poultry management practices that they see in their neighbour's homestead (preferably a poultry farm).
- Learners will be able to write an accurate, descriptive paragraph of the observations made in a logical, coherent, and clear way.

- Learners will be able to incorporate suggestions made by their peers based on the observations made above.

Learning experience:

(40 minutes)

a) The facilitator guides learners in visiting a neighbouring homestead with chickens *(preferably a poultry farm)*. The learners carefully observe, and note the chickens' behaviour and poultry management practices. Some of the behaviours may include *sunbathing, brooding, laying eggs, pests and diseases, pecking, bullying, feeding, drinking water, roosting, and sounds (danger, crowing, greeting, warning, brooding, feeding*).
    - **Note:** Learners are to record their observations in writing, drawing, or both therefore, they should carry a pen or pencil and a notebook

(20 minutes)

b) In pairs, learners compare and contrast their observations and orally comment on each other's observations. compare and contrast observations made by their peers.

Reflection Question:

1. What did you enjoy the most about the farm visit?
2. Did this activity help you learn more about chicken behaviour and management practices at home? List some of the things you observed during the visit

Way forward…

At home, together with their friends and families, learners brainstorm on why it's important to observe chicken behaviours and poultry management practices.

Reference for behaviours and physical manifestations of diseases on chicken?

Characteristic	Healthy birds	Diseased birds
Stance	• Erect • Tail held high	• Tail and wings 'dropped' • Head held close to body or twisted over back or between legs

Head	• Clean comb and wattles • Bright about the eye • Clean nostrils	• Discoloured • Shrunken comb or eyes dull or watery • Nostrils caked • Face shrunken or swollen
Muscles	• Bird feels 'solid' • Struggles vigorously when held	• Loss of weight and strength • Uneven size of thighs • Keel protrudes under skin
Legs and feet	• Clean waxy scales • Smooth joints • Cool to touch	• Dehydrated with prominent tendons • Enlarged • Warm to touch • Cracked feet
Feathers	• Smooth and neat • Shiny	• Fluffed out • Stained in abdomen area
Colour	• Breed and strain characteristics	• Less colourful • In hens excess yellow may be from reduced laying rate
Appetite and thirst	• Eat and drink water often	• Lose appetite • Drink water excessively

Droppings	• Grey • Brown with white caps • Definite form • Cecal droppings may be frothy	• White • Green • Red • Yellow • Very watery or sticky • https://the-chicken-chick.com/whats-scoop-on-chicken-poop-digestive
Abdomen	• Firm to touch • Fat birds may feel hard	• May be very hard or very soft
Vent	• Clean • Level with body surface	• Inflamed around vent area • Pasted over with droppings • Protrusion of tissues
Breathing	• Silent • Beak closed (in hot weather birds may breath with mouth open)	• Coughing • Rattling • Snickering • Obvious panting movements

DATA ANALYSIS, VISUALIZATION, & INTERPRETATION



DATA ANALYSIS, VISUALIZATION AND INTERPRETATION

Description: To come up with useful and meaningful information from data or observations, analysis has to be done. Information from the analysis is vital in making informed decisions. The graphical representation of the information and data is referred to as data visualisation.

This unit aims at building the student's ability to represent and understand real-world problems using charts, graphs and tables. Learners get to notice patterns in the data and understand the trends. In the process, learners are to analyse and visualise the chicken behaviours and poultry management practices they have observed. They discuss different charts, graphs, and tables they know, analyse and visualise the data, and then present the data.

Skills acquired in this unit cut across different subjects and the lessons can be linked to the CBC learning strands and substrands listed below;

- Mathematics, Strand 4.0 *(Data handling)*, Sub strand 4.1 *(Bar graphs).*
- Agriculture, Strand 2.0 *(Domestic animals)*, Sub strand 2.1 *(Practices in rearing small domestic animals).*
- Design art and craft, strand 1.0 *(Picture making)*, sub-strand 1.2 (*drawing*).
- English, Strand 9.1 *(Listening and Speaking)*, Sub strand 9.1.1 *(Pronunciation and Vocabulary: Speaking Fluency).*
- English, Strand 4.2 *(Reading)*, Sub strand 4.1 *(Bar graphs).*
- Mathematics, Strand 4.0 *(Data handling)*, Sub strand 4.2.1 *(Intensive Reading: visuals).*
- Social studies, Strand 1.0 *(Natural And the Built Environments)*, Sub strand 1.3 *(Climatic regions in Eastern Africa).*

At the end of this unit, the students will understand and appreciate the role that data analysis, visualisation, and interpretation play in solving real-world problems. This lesson is important to the project since it will equip the learners with the skills to come up with appropriate interpretations of the recorded observations on chicken behaviours and poultry management practices and the connection between the observations. As a follow-up to the previous lesson, learners will gain further skills in identifying patterns and interpreting data tables and bar graphs.

Objective: Learners to gain skills that they will use in analysing and visualising the chicken behaviour and poultry management practices data they will have collected so as to come up with meaningful connections between the behaviours.

Skills acquired: Communication, collaboration, creativity, self reliance, and imagination

Key question: How can we present our data in a visual format?

Estimated time: 1 hour 20 minutes over two days.

Materials: notebook, pen, chalkboard/whiteboard, coloured chalk/ marker pen,graph paper, manila papers

Activities: This unit has been divided into two lessons;

1. Creating data tables
2. Charts and graphs
3. Data interpretation

LESSON 1: Creating data tables (40 minutes)

Learning outcomes for this lesson:

- Learners will be able to organise the observed and collected chicken behaviour data into tables.

Learning experience:

(5 minutes)

a) A few learners mention the chicken behaviour and poultry management practices they have collected using the daily observation tool.

(20 minutes)

b) In groups, learners create data columns and name them. Members of the group are then to enter the observations they made using the daily observation tool in the created tables.

(5 minutes)

c) Learners discuss the advantages and disadvantages of having the data in tables.

(10 minutes)

d) Learners suggest other ways that the data can be presented. Some of the expected answers could include *bar graphs, pie charts, stem and leaf, tree diagram, tallying, etc.*

Reflection question

1. Have you learnt anything new today? /What have you learnt today?
2. What did you find interesting about organising information in tables?

LESSON 2: Charts and graphs (40 minutes)

Learning outcomes for this lesson:

- Learners will be able to present data using charts and graphs.

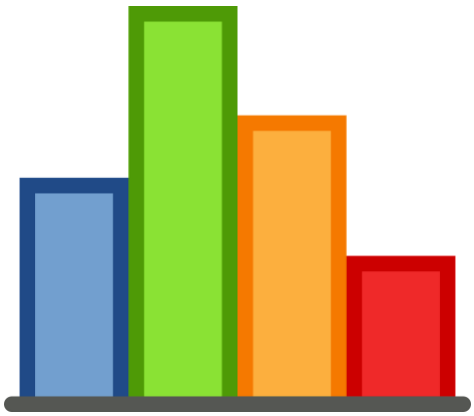
Learning experience:

(15 minutes)

a) As a class, learners list and discuss different types of charts and graphs they know and instances they have used them. Facilitator to urge the learners to take note of any other charts and graphs their peers are aware of, apart from the pie chart and bar graph.



Pie chart



Bar graph

(25 minutes)

b) In groups, the facilitator guides learners in selecting few chicken behaviours or poultry management practices recorded in the daily observation tool to be used in creating the charts and graphs. Each group presents to the rest of the class the chart or graph they have come up with. The rest of the class comments on the charts and graphs presented by the groups.

Reflection question

1. Is there anything interesting that you have learnt today?
2. As a learner, which graph did you find easy to use? Why
3. How do you think working as a group has helped you understand charts and graphs?

LESSON 3: Data interpretation (40 minutes)

Learning outcomes for this activity:

- Learners recall the different charts and graphs they learned in the previous lesson and interpret the meaning of the visualised data.

Learning experience:

(**5 minutes**)

a) As a class, learners orally mention the different charts and graphs they learned in lesson 2.

(20 minutes)

b) In groups, learners note down what they have learned from the charts and graphs they created and presented in lesson 2.

Guide questions:

- What can you learn from the chart or graph?

(15 minutes)

c) The groups present to the rest of the class what they have learned in (b) above for peer commenting. Facilitator to allow the rest of the class to ask questions and seek clarifications.

Reflection question

1. What do you think you did well during this activity?
2. How do you think working as a group has helped you understand charts and graphs?
3. Is there anything that you would do differently in the lesson today?
4. Did you encounter any challenges during this lesson? What are they?
5. How did you solve some of the challenges you encountered?
6. Can you list some of the skills that you have gained in this lesson that will help you elsewhere?

Way forward…

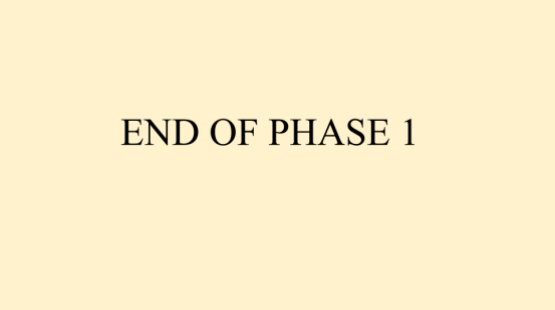
As an assignment, learners choose any recorded observation of their interest and analyse, visualise, and interpret. This can be done at home together with their friends and family.

Powerpoint Presentation

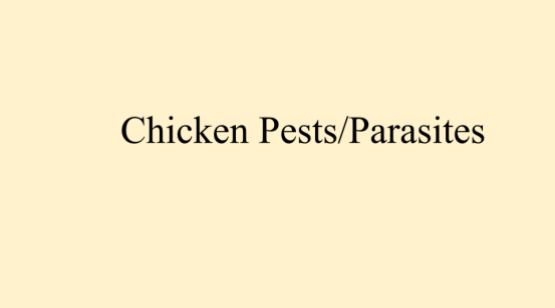
Picture story:

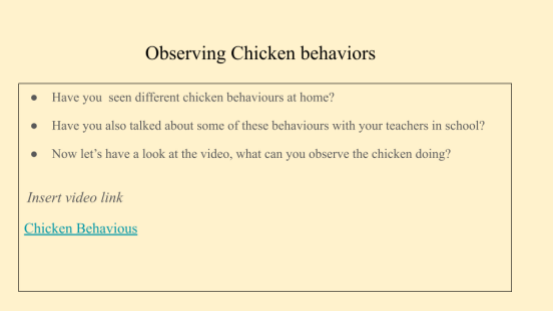
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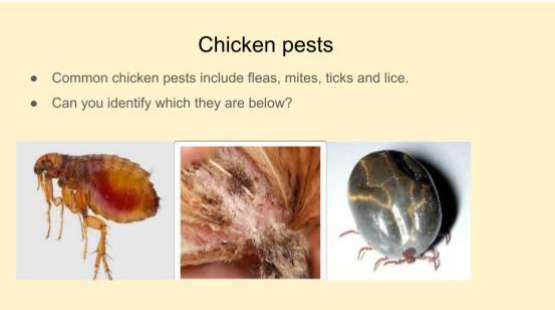


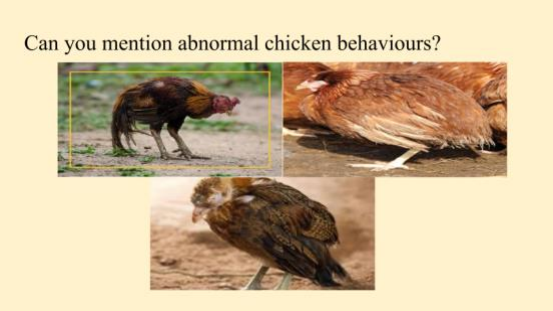


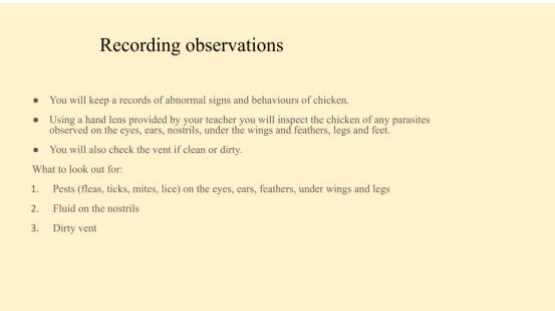












Use the pens and notebooks provided by your
To keep a record of what your observe.

Journal-



Hand lenses

Use to observe small things like mites an fleas in chicken



The End